

ZIMSEC Pure Mathematics: Group Theory

Solutions

Section A: Formal Proofs and Axioms

Question 1: Diagonal Matrix Group

(a) **Closure:** Let $A, B \in M$ where $A = \begin{bmatrix} x_1 & 0 \\ 0 & y_1 \end{bmatrix}$ and $B = \begin{bmatrix} x_2 & 0 \\ 0 & y_2 \end{bmatrix}$ with $x_1, y_1, x_2, y_2 \in \mathbb{R} \setminus \{0\}$.

Product $AB = \begin{bmatrix} x_1 \cdot x_2 & 0 \\ 0 & y_1 \cdot y_2 \end{bmatrix}$.

Since the set of non-zero real numbers is closed under multiplication, $x_1 \cdot x_2 \neq 0$ and $y_1 \cdot y_2 \neq 0$. Thus, $AB \in M$.

(b) **Associativity:** Matrix multiplication is generally associative. For $A, B, C \in M$:

$$(AB)C = \begin{bmatrix} x_1 x_2 & 0 \\ 0 & y_1 y_2 \end{bmatrix} \begin{bmatrix} x_3 & 0 \\ 0 & y_3 \end{bmatrix} = \begin{bmatrix} x_1 x_2 x_3 & 0 \\ 0 & y_1 y_2 y_3 \end{bmatrix}$$

$$A(BC) = \begin{bmatrix} x_1 & 0 \\ 0 & y_1 \end{bmatrix} \begin{bmatrix} x_2 x_3 & 0 \\ 0 & y_2 y_3 \end{bmatrix} = \begin{bmatrix} x_1 x_2 x_3 & 0 \\ 0 & y_1 y_2 y_3 \end{bmatrix}$$

Since $(AB)C = A(BC)$, associativity holds.

(c) **Identity Element:** The identity matrix I must satisfy $AI = A$.

$$\begin{bmatrix} x & 0 \\ 0 & y \end{bmatrix} \begin{bmatrix} i_1 & 0 \\ 0 & i_2 \end{bmatrix} = \begin{bmatrix} x & 0 \\ 0 & y \end{bmatrix} \Rightarrow xi_1 = x \text{ and } yi_2 = y.$$

Thus, $i_1 = 1$ and $i_2 = 1$. The identity is $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

Since $I \in \mathbb{R} \setminus \{0\}$, $I \in M$.

(d) **Inverse:** For $A = \begin{bmatrix} x & 0 \\ 0 & y \end{bmatrix}$, the inverse A^{-1} must satisfy $AA^{-1} = I$.

$$\begin{bmatrix} x & 0 \\ 0 & y \end{bmatrix} \begin{bmatrix} x' & 0 \\ 0 & y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow xx' = 1 \text{ and } yy' = 1.$$

So, $x' = 1/x$ and $y' = 1/y$.

$$A^{-1} = \begin{bmatrix} 1/x & 0 \\ 0 & 1/y \end{bmatrix}.$$

Since $x, y \neq 0$, $1/x$ and $1/y$ are defined and non-zero, so $A^{-1} \in M$.

Question 2: Multiplication Modulo 10

(a) Cayley Table for $S = \{2, 4, 6, 8\}$ under $\times_{10}$:

$\times_{10}$	2	4	6	8
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2	4	8	2	6
4	8	6	4	2
6	2	4	6	8
8	6	2	8	4

(b) Identity: Looking at the table, the row and column for **6** are identical to the header row/column. $2 \times 6 = 2$, $4 \times 6 = 4$, $6 \times 6 = 6$, $8 \times 6 = 8$. Thus, the identity element is **6**.

(c) Inverses: (Find elements that multiply to give the identity 6)

Inverse of 2: **8** ($2 \times 8 = 16 \equiv 6$)

Inverse of 4: **4** ($4 \times 4 = 16 \equiv 6$)

Inverse of 6: **6** ($6 \times 6 = 36 \equiv 6$)

Inverse of 8: **2** ($8 \times 2 = 16 \equiv 6$)

Section B: Subgroups and Cyclic Structures

Question 3: Subgroup of $(\mathbb{Z}_{12}, +_{12})$

(a) Subgroup generated by {3}:

$\langle 3 \rangle = \{3(1), 3(2), 3(3), 3(4)\} \bmod 12 = \{3, 6, 9, 0\}$.

(b) Order: The number of elements in the subgroup is **4**.

(c) Elements of order 4: An element **a** has order 4 if $4a \equiv 0 \bmod 12$ and $ka \neq 0$ for $k < 4$.

Checking elements: **3** (3, 6, 9, 0) and **9** (9, 18 $\equiv$ 6, 27 $\equiv$ 3, 36 $\equiv$ 0).

Elements are **{3, 9}**.

(d) Lagrange's Theorem: Lagrange's Theorem states that the order of a subgroup must divide the order of the group.

Order of $G = 12$. The divisors of 12 are 1, 2, 3, 4, 6, 12.

Since 5 is not a divisor of 12, there can be no subgroup of order 5.

Question 4: Prime Order Groups

(a) Proof G is cyclic: Let $a \in G$ such that $a \neq e$. Let $H = \langle a \rangle$ be the subgroup generated by a . By Lagrange's Theorem, the order of H , $|H|$, must divide $|G| = p$. Since p is prime, the only divisors are 1 and p . Since $a \neq e$, $|H| > 1$, therefore $|H| = p$. Since $H \subseteq G$ and they have the same finite order, $H = G$. Thus G is cyclic.

(b) Generator: As shown above, the subgroup generated by any non-identity element a must have an order that divides p . Because p is prime and $|\langle a \rangle| \neq 1$, it must be that $|\langle a \rangle| = p$, making a a generator.

Section C: Complex Group Proofs

Question 5: Group Properties

(a) Uniqueness of Identity: Suppose there are two identity elements e_1 and e_2 in G .

By definition of e_1 : $e_1 * e_2 = e_2$.

By definition of e_2 : $e_1 * e_2 = e_1$.

Therefore, $e_1 = e_2$. The identity is unique.

(b) Socks-and-Shoes Property: To prove $(a * b)^{-1} = b^{-1} * a^{-1}$, we show that $(a * b) * (b^{-1} * a^{-1}) = e$:

$$\begin{aligned} & (a * b) * (b^{-1} * a^{-1}) \\ &= a * (b * b^{-1}) * a^{-1} \text{ (by associativity)} \\ &= a * e * a^{-1} \text{ (by inverse property)} \\ &= a * a^{-1} \text{ (by identity property)} \\ &= e \text{ (by inverse property)} \end{aligned}$$

Similarly, $(b^{-1} * a^{-1}) * (a * b) = e$. Thus, $b^{-1} * a^{-1}$ is the inverse of $a * b$.